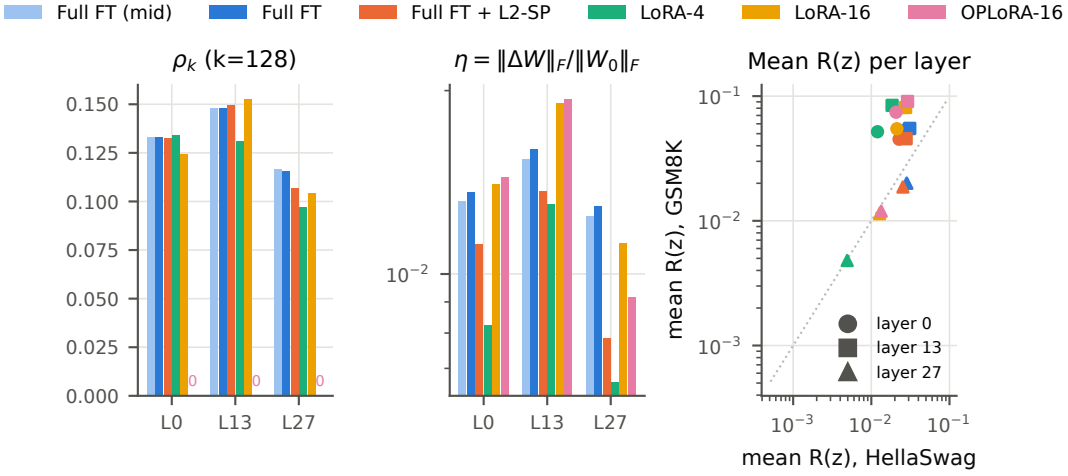
Full FT (mid)
Full FT
Full FT + L2-SP
LoRA-4
LoRA-16
OPLoRA-16
ρ_k (k=128)
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
0
0
L0
L13
L27
$\eta = \|\Delta W\|_F / \|W_0\|_F$
10^{-2}
L0
L13
L27
Mean R(z) per layer
10^{-1}
10^{-2}
10^{-3}
mean R(z), GSM8K
layer 0
layer 13
layer 27
10^{-3}
10^{-2}
10^{-1}
mean R(z), HellaSwag